

A Hierarchical Multimodal Ensemble with FiLM-Conditioned Fusion for Tongue and Face Age Estimation

1st Place Solution · IJCB 2026 MBAge Challenge (CodaBench Competition #15116)

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Abstract

We present the winning solution to the IJCB 2026 MBAge challenge, in which we achieved first place on the final combined leaderboard ($S_{\text{final}} = 0.4129$) with the top Phase 2 score of $S_2 = 0.4396$. Our approach treats age estimation as a structured prediction problem and combines three complementary deep models trained on pre-extracted tongue (four CNN backbones) and face (Gabor magnitude banks) features: a hierarchical decade-then-fine Deep Label Distribution Learning (DLDL) model, a feature-engineered dual-backbone model incorporating fourteen statistical and attention-based descriptors, and a Feature-wise Linear Modulation (FiLM) ensemble that median-aggregates six seed-varied face-conditioned predictors. A weighted ensemble of the three components, with weights refined by a 114-candidate grid search constrained to out-of-fold non-regression, yielded our final submission. Code, predictions, and reproducibility artifacts are publicly released.

Method

Hierarchical decade-routed DLDL (40% weight). The first component decomposes regression into a five-way decade classification (20s–60s) followed by ten-bin within-decade Deep Label Distribution Learning [1]. A shared three-layer MLP trunk (256 hidden units, GELU, LayerNorm, dropout 0.3) processes global-average-pooled features from four CNN backbones (ResNet18, MobileNetV2, EfficientNetB0, DenseNet121) concatenated with Gabor face features (cheek and chin banks B/C/D, each 40-D). Predictions follow soft hierarchical routing,

$$\hat{y} = \sum_{d=1}^5 P(d) \left(\text{start}_d + \sum_{k=0}^9 P(k | d) \cdot k \right).$$

Training combines decade cross-entropy, within-decade KL divergence against Gaussian soft labels (σ annealed $2.0 \rightarrow 0.5$), L_1 regression, and Wing loss [2], optimized with AdamW under ten-fold stratified cross-validation (200 epochs, patience 30).

Feature-engineered dual-backbone model (42.5% weight). The second component is a DenseNet121 + ResNet18 dual-backbone hierarchical model (hidden dim 384, dropout 0.35) augmented with all fourteen of the following: pre-pool LayerNorm; standard-deviation and 2×2 grid pooling; discrete-cosine and Haar wavelet projections; higher-order and spatial moments; Gabor-bank marginals and cross-bank asymmetry; iterative-square-root covariance pooling [3] (reduce-dim 32); Compact Bilinear Pooling [4] (dim 1024); mutual-information channel selection (top-64); Efficient Channel Attention [5]; CKA-based backbone de-duplication; learned-query attention pooling; confidence-weighted late fusion; and the Multi-Modal Transfer Module (MMTM) [6] for cross-backbone fusion.

FiLM-conditioned median ensemble (17.5% weight). The third component is a seven-model median over six Feature-wise Linear Modulation [7] transformers ($d_{\text{model}} = 128$, four heads, two layers, triple-head decoder combining DLDL, regression, and ordinal predictions) trained with seeds $\{0, 1, 2, 3, 4, 42\}$, plus the hierarchical model. FiLM projects face features into per-channel affine parameters γ, β that modulate the tongue representation. Median aggregation is robust to single-seed outliers and reduces prediction variance relative to arithmetic averaging.

Final blend. Component predictions were combined as a weighted average and clipped to the valid age range:

$$\hat{y} = \text{clip}(0.400 \hat{y}_{\text{hier}} + 0.425 \hat{y}_{\text{s12}} + 0.175 \hat{y}_{\text{film6}}, 20, 69).$$

Weights were selected by exhaustive grid search at 0.025 resolution over the ternary simplex (114 candidate combinations), retaining the smallest deviation from the reference (40/40/20) weighting that yielded a positive out-of-fold jittered S delta with no negative per-metric component. The refinement contributed $+0.00022$ to the final leaderboard score over the equal-weight reference, validating the proven blend structure as a tight local optimum.

Results

Configuration	Rank	S	MSE	1Y-ACC	5Y-ACC	10Y-ACC
Phase 1 (tongue only)	2nd	0.3727	97.17	0.16	0.53	0.74
Phase 2 (tongue + face)	1st	0.4396	46.92	0.176	0.648	0.864
Combined ($0.4 S_1 + 0.6 S_2$)	1st	0.4129	—	—	—	—

Table 1. Final CodaBench leaderboard metrics. The competition score is computed as $S = 0.1/(1 + \text{MSE}) + 0.4 \text{ACC}_{1Y} + 0.3 \text{ACC}_{5Y} + 0.2 \text{ACC}_{10Y}$. Our team (ks7585) won Phase 2 outright and the combined ranking by a margin of $+0.0027$ over the second-place team.

Reproducibility and Code Availability

All eight component models were trained on a single NVIDIA RTX 5070 (12 GB) using PyTorch 2.11.0+cu128 and Python 3.12, in a total wall-clock time of approximately 3.5 hours. Reproducibility was verified by retraining the hierarchical model from scratch on different hardware and observing byte-identical out-of-fold predictions to the original training run (MD5 9ee818c7a9557ea0852ff9634bd62227), confirming full determinism of the recipe under fixed random seeds. The complete code base, including all training scripts, out-of-fold predictions, the final submission file, and step-by-step reproduction instructions, is released at github.com/ks7585/MBAge-IJCB2026 (MIT License).

Key References

[1] B.-B. Gao *et al.*, “Deep Label Distribution Learning with Label Ambiguity,” *IEEE Trans. Image Processing*, 26(6):2825–2838, 2017. [2] Z.-H. Feng *et al.*, “Wing Loss for Robust Facial Landmark Localisation,” *CVPR*, 2018. [3] P. Li *et al.*, “Towards Faster Training of Global Covariance Pooling Networks by Iterative Matrix Square Root Normalization,” *CVPR*, 2018. [4] Y. Gao *et al.*, “Compact Bilinear Pooling,” *CVPR*, 2016. [5] Q. Wang *et al.*, “ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks,” *CVPR*, 2020. [6] H. R. V. Joze *et al.*, “MMTM: Multimodal Transfer Module for CNN Fusion,” *CVPR*, 2020. [7] E. Perez *et al.*, “FiLM: Visual Reasoning with a General Conditioning Layer,” *AAAI*, 2018.